

Recursive Least Squares

Model

$$y_i = C_i x + e_i, \quad i=1, 2, \dots$$

$$y_i \in \mathbb{R}^m, \quad x \in \mathbb{R}^n, \quad e_i \in \mathbb{R}^m, \quad C_i = m \times n \text{ real.}$$

i - represents time index

x - an unknown constant vector

y_i = measurements

e_i = model "mismatch"

Objective 1: Compute a least squared error estimate of x at time k , using all available data,

$$y_1, \dots, y_k \quad \triangleright$$

Objective 2 Discover a computationally attractive form for the answer.

Solution

$$\begin{aligned} \hat{x}_k &:= \arg \min_x \left(\sum_{i=1}^k (y_i - c_i x)^2 S_i (y_i - c_i x) \right) \\ &= \arg \min_x \left(\sum_{i=1}^k e_i^T S_i e_i \right) \end{aligned}$$

where $S_i = \text{mxm}$ positive def. matrix

$$(S_i > 0)$$

$$Y_k := \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix}, \quad A_k = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{bmatrix}, \quad E_k = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_k \end{bmatrix}$$

$$R_k = \text{diag}(S_1, S_2, \dots, S_k) > 0$$

$$Y_k = A_k x + E_k$$

$$\|Y_k - A_k x\|^2 = \|E_k\|^2 := E_k^T R_k E_k$$

$\hat{x}_k := \arg \min_x \|Y_k - A_k x\|$ satisfies

the Normal Equations

$$(A_k^T R_k A_k) \hat{x}_k = A_k^T R_k Y_k$$

$$\therefore \hat{x}_k = (A_k^T R_k A_k)^{-1} A_k^T R_k Y_k$$

This is called a Batch Solution.

Drawback $A_k^T R_k A_k = km \times km$ matrix, and grows at each step!

Solution Find a recursive means to compute $\hat{x}_{k+1}$ in terms of $\hat{x}_k$ and the new measurement y_{k+1} ∇_0

Normal equation at time k

$$A_k^T R_k A_k \hat{x}_k = A_k^T Y_k$$

is equivalent to

$$\left(\sum_{i=1}^k C_i^T S_i C_i \right) \hat{x}_k = \sum_{i=1}^k C_i^T S_i y_i$$

We define

$$P_k = \sum_{i=1}^k C_i^T S_i C_i$$

so that

$$P_{k+1} = P_k + C_{k+1}^T S_{k+1} C_{k+1}$$

At time $k+1$,

$$\left(\sum_{i=1}^{k+1} c_i^T s_i c_i \right) \hat{x}_{k+1} = \sum_{i=1}^{k+1} c_i^T s_i y_i$$

OR

$$Q_{k+1} \hat{x}_{k+1} = \underbrace{\sum_{i=1}^k c_i^T s_i y_i}_{Q_k \hat{x}_k} + C_{k+1}^T s_{k+1} y_{k+1}$$

$$\therefore \boxed{Q_{k+1} \hat{x}_{k+1} = Q_k \hat{x}_k + C_{k+1}^T s_{k+1} y_{k+1}}$$

Good start!

Estimate at time $k+1$ expressed as a linear combination of the estimate at time k and the latest measurement at time $k+1$.

Continuing,

$$\hat{x}_{k+1} = P_{k+1}^{-1} [P_k \hat{x}_k + C_{k+1} S_{k+1} y_{k+1}]$$

Because $P_k = P_{k+1} - C_{k+1}^T S_{k+1} C_{k+1}$,

we have

$$\hat{x}_{k+1} = \hat{x}_k + \underbrace{P_{k+1}^{-1} C_{k+1}^T S_{k+1}}_{\text{Kalman Gain}} \underbrace{(y_{k+1} - C_{k+1} \hat{x}_k)}_{\text{Innovations}}$$

Innovations $y_{k+1} - C_{k+1} \hat{x}_k$ = Measurement
at time $k+1$ minus the "predicted" value
of the measurement = "new information".

In a real-time implementation, computing the inverse of Q_{k+1} can be time consuming. An attractive alternative can be avoided by applying the Matrix Inversion Lemma:

$$Q_{k+1}^{-1} = (Q_k + C_{k+1}^T S_{k+1} C_{k+1})^{-1}$$

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(DA^{-1}B + C^{-1})^{-1}DA^{-1}$$

$$A \leftrightarrow Q_k \quad B \leftrightarrow C_{k+1}^T \quad C \leftrightarrow S_{k+1} \quad D \leftrightarrow C_{k+1}$$

$$Q_{k+1}^{-1} = Q_k^{-1} - Q_k^{-1} C_{k+1}^T [C_{k+1} Q_k^{-1} C_{k+1}^T + S_{k+1}^{-1}]^{-1} C_{k+1} Q_k^{-1}$$

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Which is a recursion for Q_k^{-1} !

Upon defining

$$P_k = Q_k^{-1}$$

we have

$$P_{k+1} = P_k - P_k C_{k+1}^T [C_{k+1} P_k C_{k+1}^T + S_{k+1}^{-1}]^{-1} C_{k+1} P_k$$

We note that we are now inverting a matrix that is $m \times m$, instead of one that is $n \times n$. Typically, $n > m$, sometimes by a lot !